



Exploration of overloaded cation exchange chromatography for monoclonal antibody purification

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ABSTRACT

Cation exchange chromatography using conventional resins, having either diffusive or perfusive flow paths, operated in bind-elute mode has been commonly employed in monoclonal antibody (MAb) purification processes. In this study, the performance of diffusive and perfusive cation exchange resins (SP-Sepharose FF (SPSFF) and Poros 50HS) and a convective cation exchange membrane (Mustang S) and monolith (SO₃ Monolith) were compared. All matrices were utilized in an isocratic state under typical binding conditions with an antibody load of up to 1000 g/L of chromatographic matrix. The dynamic binding capacity of the cation exchange resins is typically below 100 g/L resin, so they were loaded beyond the point of anticipated MAb break through. All of the matrices performed similarly in that they effectively retained host cell protein and DNA during the loading and wash steps, while antibody flowed through each matrix after its dynamic binding capacity was reached. The matrices differed, though, in that conventional diffusive and perfusive chromatographic resins (SPSFF and Poros 50HS) demonstrated a higher binding capacity for high molecular weight species (HMW) than convective flow matrices (membrane and monolith); Poros 50HS displayed the highest HMW binding capacity. Further exploration of the conventional chromatographic resins in an isocratic overloaded mode demonstrated that the impurity binding capacity was well maintained on Poros 50HS, but not on SPSFF, when the operating flow rate was as high as 36 column volumes per hour. Host cell protein and HMW removal by Poros 50HS was affected by altering the loading conductivity. A higher percentage of host cell protein removal was achieved at a low conductivity of 3 mS/cm. HMW binding capacity was optimized at 5 mS/cm. Our data from runs on Poros 50HS resin also showed that leached protein A and cell culture additive such as gentamicin were able to be removed under the isocratic overloaded condition. Lastly, a MAb purification process employing protein A affinity chromatography, isocratic overloaded cation exchange chromatography using Poros 50HS and anion exchange chromatography using QSFF in flow through mode was compared with the MAb's commercial manufacturing process, which consisted of protein A affinity chromatography, cation exchange chromatography using SPSFF in bind-elute mode and anion exchange chromatography using QSFF in flow through mode. Comparable step yield and impurity clearance were obtained by the two processes.

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1. Introduction

At present, there are hundreds of monoclonal antibodies (MAbs) that are either currently on the market or are under development

[1]. MAbs are derived from a common framework and have a high degree of homology, which results in similar properties. These similarities are being exploited as companies establish platform processes that allow for the purification of similar molecules with minimal development and optimization, especially for early clinical stage development. As a project moves to late clinical development and commercial stages, however, further development is typically required to address cell culture process changes, increased antibody titer, and facility fit, as well as to optimize the yield and reduce the cost for the commercial process.

Downstream processing of MAbs is increasingly centering on a platform process that includes Protein A affinity

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chromatography as a capture step, followed by one or two polishing steps such as ion exchange, hydrophobic interaction, or mixed mode chromatography. Host cell protein, host cell DNA, high molecular weight species, and other process-related impurities can usually be reduced to levels low enough to be considered acceptable by the FDA by this standard approach. Reduction of endogenous and adventitious viruses can also be achieved [2–6].

These classic purification processes have typically been operated in bind-elute mode for protein A, cation exchange, hydrophobic interaction and mixed mode chromatography, while anion exchange has mostly been employed in flow through mode. One approach to reducing the downstream cost of purifying monoclonal antibodies from CHO derived cell culture fluid has been to leverage recent developments in anion exchange chromatography, including weak partitioning or partial binding chromatography, to enable a two column purification process (using protein A and anion exchange, followed by a viral filtration step) [7,8]. Throughput with anion exchange chromatography can reach >250 g/L of column volume; however, success with the two column process is limited when faced with a higher level of high molecular weight species in the feedstock or an antibody with a high pI value (≥ 9). Removing residual process related impurities that are added or generated during cell culture processes can also be a challenge for the two column process since these impurities can be more effectively removed by cation exchange or other types of chromatography. Therefore, a three column process is still widely used to purify monoclonal antibodies.

In the search for ways to simplify and reduce the cost of the traditional three column approach, ion exchange membranes and monoliths have also been explored [9–11]. Recent discoveries with anion and cation exchange membranes used as the second or third purification step to effectively remove product and process related impurities have been reported [9,12,13]. Membrane chromatography provides an advantage over traditional column chromatography because convective mass transfer dominates in the flow path. Monolithic matrices may also enable a fluid flow that is dominated by convective mass transfer with a minor component of diffusion. Convective mass transport is more efficient, resulting in a dynamic binding capacity (DBC) that is mostly independent of flow rate. Effective resolution and separation power across a broad range of flow rates is maintained, allowing for considerable process efficiency improvements [10,14–16]. Membranes and monoliths can be used as single- or multi-use, disposable matrices. Lower buffer volume consumption, eliminating the need for packing and unpacking large columns, and reducing the complexity and size of a plant footprint are also advantages with membrane and monolith matrices.

The original intent of this work was to further explore the use of cationic monolithic matrices and membranes as an alternative to the traditional cation exchange chromatography step. Overloading of a traditional cationic chromatography column was incorporated into the experimental design for the purpose of understanding and comparing the binding of antibodies and impurities to various matrices. To that end, four cation exchange chromatographic matrices having diffusive, perfusive, and convective flow properties were selected for comparison. An isocratic running condition was used under typical binding conditions for monoclonal antibodies with an antibody loading amount of up to 1000 g/L of chromatographic matrix. This loading density clearly exceeded the point of MAb breakthrough, but was used to fully explore the binding of impurities, which may reach breakthrough at a higher loading value. The most intriguing result of this work was the performance of the traditional cationic chromatography column. Operating these columns in an isocratic overloaded mode demonstrated promise to achieve a more efficient and/or cost effective approach compared to standard bind-elute cationic chro-

matography, while still providing acceptable levels of impurity removal.

Based on this observation, further studies were performed with overloaded conditions to explore the influence of processing parameters, such as flow rate and conductivity, on the removal of impurities. Finally, isocratic overloaded cation exchange chromatography was incorporated into a MAb purification process, where the performance was compared to the traditional bind-elute process. Superior processing performance was demonstrated with isocratic overloaded mode, even under heavily overloaded conditions. While this process used standard cation exchange resin, performing the process in an isocratic overloaded mode provided a means of reducing the resin volume, equipment size, processing time and process complexity required for a given amount of product, thus offering the potential to lower the cost of the purification process.

2. Materials and methods

2.1. Materials

Two types of small scale cation exchange membranes were tested, the Mustang S and the Sartobind S. The Mustang S is a PES membrane coupled with sulfonfyl ligand sold by Pall Corporation (Port Washington, NY). Two different sizes of this filter were used in the study: the Acrodisc S is their syringe scale filter (0.18 mL of the membrane bed volume, 6 layers) and the Mustang Coin unit is their coin filter S (0.35 mL of the membrane bed volume, 16 layers). A 13 mm O.D. (10 mm I.D.) stainless steel membrane holder from Pall Corporation was used with the coin filter. Sartobind S is another PES membrane coupled with sulfonfyl ligand (1 mL of the membrane volume, 15 layers), and it was purchased from Sartorius Corporation (Goettingen, Germany). The monolithic SO₃ disks tested were coupled with sulfonfyl ligand (0.34 mL of monolith volume per disk, 13 mm I.D. $\times$ 2 mm thickness). The monolithic discs and disk holder were purchased from BIA Separation (Villach, Austria). SP-Sepharose Fast Flow (SPSFF) was bought from G.E. Healthcare (Uppsala, Sweden). SPSFF is a cation exchange resin coupled with sulphopropyl ligand; the resin has a particle size range of 80–100 μ m. Poros 50HS is a cation exchange resin coupled with sulphopropyl ligand and was purchased from Applied Biosystems (Foster City, CA). The average particle size of Poros 50HS resin is 50 μ m. Small scale glass column hardware with 6.6 mm I.D., obtained from Omnifit (Diba Industries, Mahopac, NY), was used to conduct the SPSFF and Poros 50HS column chromatography experiments.

The load material for this study was Chinese hamster ovary (CHO) cell derived cell culture fluid that was generated and subsequently purified through protein A chromatography at Genentech's Oceanside facility (Oceanside, CA, USA). The cell culture fluids containing each monoclonal antibody were harvested by continuous centrifuge to remove cells and cellular debris, and further clarified through filtration with depth filters and a 0.2 μ m filter. For the protein A chromatography step, the loading and washing conditions were at neutral pH, and elution was performed at low pH. The protein A protein pools were adjusted to the appropriate pH and conductivity for loading onto cation exchange chromatography, followed by 0.22 μ m filtration. The same pH and conductivity were used for the cation exchange chromatography loading, equilibration, and post-load wash conditions. Concentration of product (g/L) in the protein A and cation exchange pools was determined by absorbance (278 nm) on a UV–Vis spectrophotometer. All chromatography runs performed with the membrane, monolith, and resin particle packed columns were run on an ÄKTA Explorer 100 system from G.E. Healthcare.

Three humanized monoclonal IgG1 antibodies were utilized in this study. The antibodies have pIs between 8.1 and 9.1, are glycosylated, and bind strongly on cation exchangers in the acidic pH range at optimized conductivities.

2.2. Chromatography method

For the membrane and monolithic chromatography, the flow rate used was 100–400 membrane or monolith volumes (MV) per hour, where membrane or monolith volume is defined as the total volume held by the membrane or monolith in their holders. For the resin particle packed column chromatography, the column dimensions were 0.66 cm I.D. by 5–15 cm bed height. The flow rates used were 50–200 cm per hour (9–36 column volume (CV) per hour). All chromatography was monitored at 280 nm and performed at room temperature.

To determine the DBC of each cation exchange matrix, the protein pool was loaded at a pre-defined flow rate until reaching or exceeding the binding breakthrough at 5% of protein concentration in the load.

The loading condition used in this study was selected based on previous knowledge and performance data of monoclonal antibodies on cation exchange resins [2,17–19]. To avoid exclusion of antibody from the chromatography resin, for example, the optimized loading conductivity should be 4–6 mS/cm at pH 5–6, under which the maximum DBC could be obtained. The conductivity of 5 mS/cm and pH 5.5 were used as the loading conditions on the cation exchange matrices in the initial study to compare the performance of chromatographic matrices having convective and diffusive mass transfer properties.

For the chromatography runs performed in the initial study, monoclonal antibodies were loaded up to 1000 g/L of the chromatographic matrix volume (column volume (CV) or membrane volume or monolith volume (MV)) during each run. A wash step using the matrix equilibration buffer was applied immediately following the loading step, until the UV trace returned to baseline. A strip step with a higher salt concentration solution was used to elute the bound species from the matrix during each chromatographic run. Columns and monoliths were cleaned with 0.5 N NaOH solution after each run and stored in 0.1 N NaOH solution. In a subsequent set of experiments, the loading conditions were altered by varying the flow rate (18 and 36 CV/h) for the Poros 50HS and SPSFF column chromatography processes and by varying the load conductivity (3, 5, 8, 13 and 18 mS/cm) for the Poros 50HS column chromatography only. Otherwise, the same procedures were followed. During the loading and washing steps for each experiment, the flow-through from each run was fractionated. The protein concentration in each fraction was measured using a UV–Vis spectrophotometer. The impurity contents such as CHO host cell protein (CHOP), DNA, leached protein A, and gentamicin were determined in selected fractions.

2.3. Assays

The antibody concentration in harvested cell culture fluid (HCCF) was determined using a Poros Protein A HPLC assay (Applied Biosystems, Foster City, CA). Absorbance at 280 nm was monitored, and the elution peak area was used to quantify antibody concentration from a standard curve. Antibody concentration in purified pools was determined by absorbance at 278 nm (with absorbance at 320 nm subtracted to correct for light scattering), using a spectrophotometer with a 10 mm pathlength flow cell. The concentration was calculated as [(absorbance at 278 nm – absorbance at 320 nm) × dilution factor]/extinction coefficient.

Host cell protein (Chinese hamster ovary cell protein, or CHOP) levels were measured by an enzyme-linked immunosorbent assay

Table 1

Dynamic binding capacity of MAb1 measured at 5% of breakthrough curve.

Flow rate (CV* or MV**/h)	DBC at 5% BT (g/L CV* or MV**)
<i>SPSFF column (0.66 cm id × 5.5 cm = 1.88 mL)</i>	
9	67
18	53
36	41
55	31
<i>Poros 50HS column (0.66 cm id × 5.4 cm = 1.85 mL)</i>	
9	56
18	51
37	47
56	44
<i>Mustang S membrane (coin – 0.35 mL)</i>	
171	22
<i>Mustang S membrane (Acrodisc – 0.18 mL)</i>	
171	21
<i>Monolith SO₃ disk (0.34 mL – 1 disk)</i>	
88	18
177	18
<i>Monolith SO₃ disk (0.68 mL – 2 disk)</i>	
88	18

* CV is SPSFF or Poros 50HS column volume.

** MV is Mustang S membrane or monolith SO₃ disk volume.

(ELISA). Host cell DNA was quantified by a real-time polymerase chain reaction (TaqMan PCR) assay. Levels of leached protein A and gentamicin were also determined using ELISA-based assays. All ELISA assays used in this study were developed at Genentech.

The size heterogeneity of the monoclonal antibodies was determined by a high-performance size exclusion chromatography (SEC) assay employing a TSK-GEL G3000SWXL column (7.8 mm × 300 mm, Tosoh Bioscience, Montgomeryville, PA) to separate monomer and higher molecular weight species (HMWs). Absorbance at 280 nm was used to monitor the separation of monomer and HMW species. Percentages of monomer and HMW species were calculated based on their peak areas.

3. Results and discussion

3.1. Dynamic binding capacity (DBC)

The cation exchange ligand density was different between the chromatographic matrices. For a given ligand type, the dynamic protein binding capacity can not be predicted based only on the ligand density on the matrix. DBC is not only related to the ligand density, but also mass transport efficiency, protein charge characteristics, protein steric hindrance factor, etc. [17,20]. The effect of mass transport efficiency on the DBC of conventional chromatography matrices is pronounced.

The DBC of MAb1 was compared on each of the cation exchange matrices used in this study. The data obtained at 5% of binding break through is presented in Table 1. The capacities showed that the conventional diffusive resin, SPSFF, had the highest capacity; however, this resin also demonstrated the strongest dependence on flow rate, as its DBC decreased by over 50% as the flow rate increased from 9 to 55 CV/h. The conventional perfusive resin, Poros 50HS, demonstrated a lower capacity than SPSFF in the flow rate range of 9–18 CV/h, but a higher capacity when the flow rate was greater than 18 CV/h. The intra-particle flow path within the perfusive resin particles, and perhaps in combination with the smaller particle size of Poros 50HS, provided better mass transport and resulted in a reduced dependency of the MAb binding capacity on residence time. For the monolith SO₃, the DBC at two flow rates was measured, and as expected, no difference was observed due to its convective mass transport property. The matrix volume was also doubled while holding the flow rate constant, and this also

had no discernable impact on DBC (data not shown). Knudsen et al. [14] demonstrated that the same result was true for cationic membranes. Their study demonstrated that the breakthrough capacity of the cation exchange membrane was relatively constant over a wide range of flow rates (20–1000 MV/h). Therefore, Mustang S was evaluated at one flow rate, across two different membrane sizes. Flow rate and scale did not affect the capacity of the cationic membrane. The DBC of MAb2 was also measured at a flow rate of 18 CV/h on SPSFF and Poros 50HS and 170 MV/h on Mustang S and Monolith SO₃. MAb2 exhibited similar capacities to MAb1. A capacity of 55 g/L was obtained for SPSFF, 53 g/L for Poros 50HS, 23 g/L for Mustang S, and 21 g/L for Monolith SO₃.

3.2. Isocratic overloaded chromatography mode

Traditionally, cation exchange chromatography has been operated in bind-elute mode. Often, protein was bound to the cation exchange matrix at pH 5–6, using low conductivity, <10 mS/cm, while achieving a capacity <55 g/L. In isocratic overloaded chromatography, MAb was bound to the matrix under traditional binding conditions. However, the amount of material loaded onto the cation exchange matrix far exceeded MAb breakthrough. Once breakthrough was achieved, the process continued in isocratic mode, loading up to 1000 g/L load density. Isocratic overloaded chromatography described in this study differs from overloaded chromatography described by Mazsaroff et al. [21], where protein mass was loaded at less than 100 percent of column capacity under protein binding conditions and a gradient elution was performed to separate different proteins.

Isocratic overloaded chromatography was evaluated with MAb1 and MAb2 on different convective and diffusive matrices, including Poros 50HS, SPSFF, SO₃ monolithic disk, and a Mustang S membrane. The matrices were equilibrated, loaded, and washed under isocratic conditions, with acetate solutions at pH 5.5 and 5 mS/cm conductivity. The chromatography conditions for this experiment were chosen based on prior knowledge of the behavior of these MABs when purified using traditional bind-elute column chromatography. The cation exchange matrices were loaded up to 1000 g/L, with effluent collection beginning at UV breakthrough and ending with a post-load wash step (similar to flow-through mode).

In all cases, the both MABs initially bound to the matrix, underwent a breakthrough transitional period once the dynamic binding capacity was reached, and then continued in isocratic overloaded mode (Fig. 1). The chromatograms appeared similar to a normal flow-through chromatogram, with the exception that the formation of the MAB breakthrough curve was delayed until the MAB reached its DBC. Measurement of the MAB concentration in the continuous effluent fractions, collected throughout the experiment, exhibited a yield >90%.

3.2.1. CHOP and DNA removal

When operated in a regular bind-elute mode with a linear gradient or step-elution program, cation exchange chromatography is employed in the MAB purification process to reduce process-related impurities such as CHOP, DNA, cell culture additives, and product-related high molecular weight species. To evaluate the impurity removal by diffusive and convective matrices, the effluent fractions were tested for impurities. Figs. 2 and 3 exhibit the removal of CHOP and DNA, respectively. Approximately 70–80% of CHOP was captured by the cation exchange matrices from the MAB1 load. All matrices showed a gradual reduction in the amount of CHOP that bound as the total mass of load and mass collected increased; however, none of the matrices reached a saturation point even after 800 g of MAB1 were collected per liter of CV or MV. The monolith demonstrated a slightly higher binding capacity for CHOP

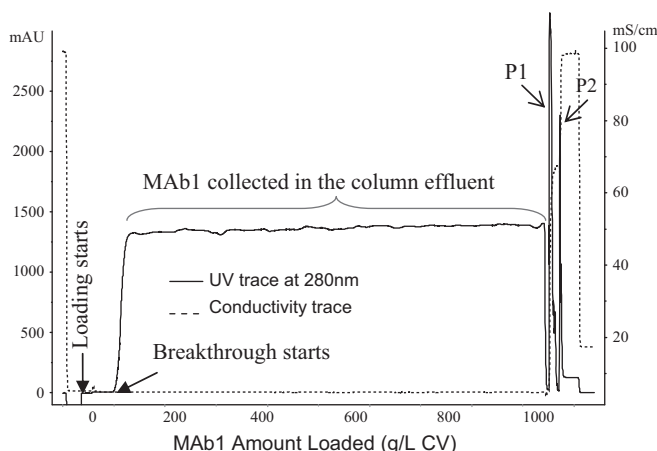


Fig. 1. MAb1 chromatogram obtained in isocratic overloaded chromatographic condition on Poros 50HS column (0.66 cm I.D. × 5.4 cm bed height) at flow rate of 18 column volume per hour. P1 is the strip peak with 350 mM sodium acetate and P2 is the cleaning peak with 0.5 N sodium hydroxide.

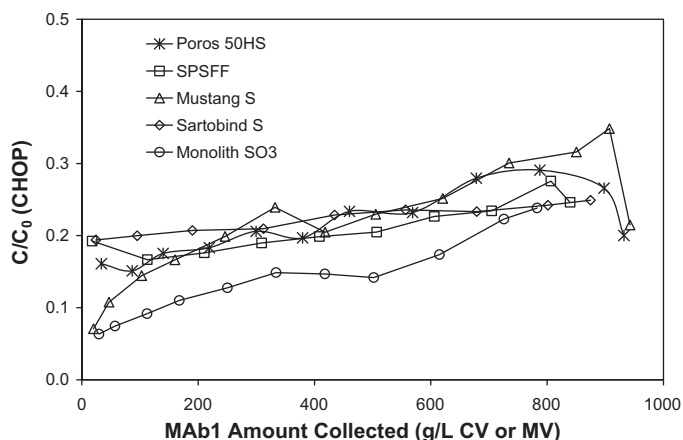


Fig. 2. Removal of host cell protein in perfusive, diffusive, and convective matrices. C/Co on y-axis is the ratio of host cell protein concentration in the effluent to that in the load. The x-axis was the cumulative MAB1 amount collected in the effluents during the isocratic overloaded runs. Note: The cumulative MAB1 amount collected was obtained through calculation of the collected column effluent volume and MAB1 concentration in each effluent.

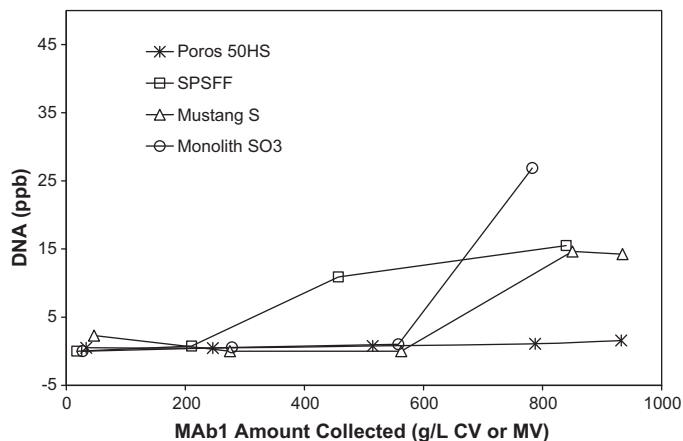


Fig. 3. Removal of host cell DNA in perfusive, diffusive, and convective matrices with MAB1. DNA concentration in the load was as high as 4000 ppb. Equilibration and wash buffer was sodium acetate at pH 5.5 and 5 mS/cm.

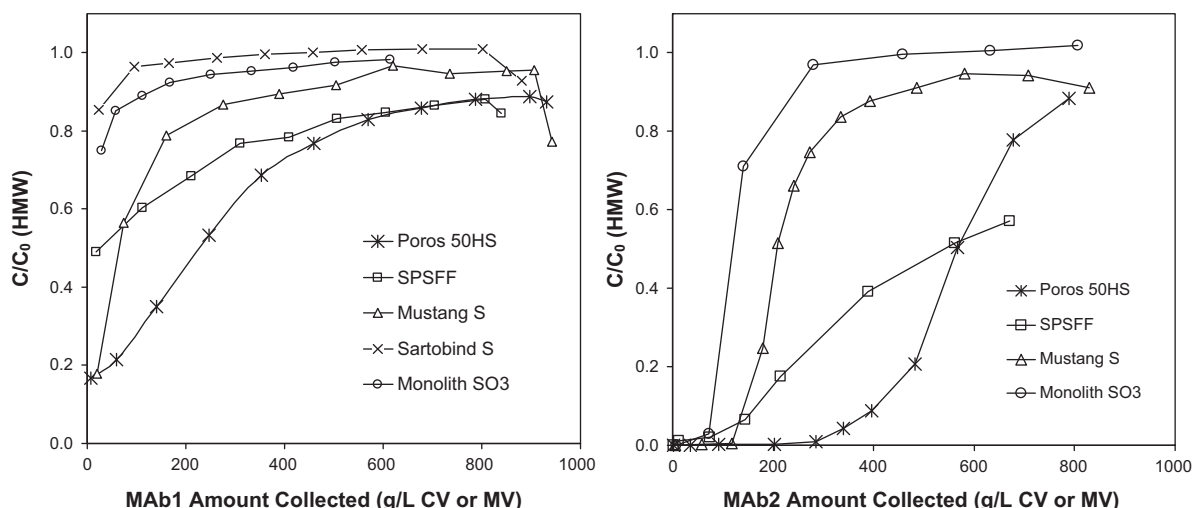


Fig. 4. Removal of high molecule weight species (HMW) by perfusive, diffusive, and convective matrices. C/C_0 is the ratio of HMW percentage in the effluent fraction to that in the load. With MAb1, HMW percentage in the load is about 5%. Flow rates were 18.5 column volumes per hour on SPSFF and Poros 50HS columns, 176 monolith volumes per hour on SO_3 monolith, and 171 membrane volumes per hour on Mustang S and Sartobind S membranes. Equilibration and wash buffer was made of sodium acetate at pH 5.5 and 5 mS/cm. With MAb2, HMW percentage in the load is about 6%. Flow rates were 15–16 column volumes per hour for SPSFF and Poros 50HS columns, 353 monolith volumes per hour on SO_3 monolith, and 333 membrane volumes per hour on Mustang S membrane. Equilibration and wash buffer was sodium acetate at pH 5.5 and 5 mS/cm.

than the other matrices. This same trend was observed for MAb2 (data not shown).

CHOP is a group of cellular proteins that vary in molecular weight, surface charge, and solubility at different pH, conductivity, and solution environments. The isoelectric points of host cell protein species cover a wide range. Some of them have a net positive charge at the load pH, which can contribute to some host cell proteins binding to the matrices. However, the net charge may not be the only factor that affects the binding affinity of proteins to ion exchange matrices; the density of the charges on the protein can also play an important role in determining the binding strength. This phenomenon has been observed with different monoclonal antibodies that have similar pI, but exhibit different binding strength (internal development data not shown here). The data generated in this study suggests that cation exchange chromatography, employed as a polishing step using the isocratic binding condition and overloading mode, can still be an effective tool for the removal of host cell protein.

Host cell DNA also shows strong binding on all the matrices under the tested condition. Fig. 3 shows that the DNA was substantially reduced (less than 40 ppb) for all the matrices, even when the MAb was loaded up to 1000 g/L of matrix. DNA enriched fractions were collected in the elution step (data not shown in this publication) to confirm that the DNA was in fact bound to the matrix. DNA is believed to be a highly negatively charged molecule at this load pH, due to the phosphate groups of the nucleotide. The mechanism of DNA binding to the cation exchange matrices is not known at this time.

3.2.2. HMW removal

High molecular weight species (HMWs) removal from MAb feed streams is typically achieved using bind-elute chromatography steps, with ion exchange, hydrophobic interaction, or hydroxyapatite resins. HMWs binds more tightly to the resin, and remains primarily bound after the product is eluted [22–25]. HMWs breakthrough curve under the overloaded condition for cation exchange is illustrated in Fig. 4, for both MAb1 and MAb2.

For MAb2, the HMWs bound from the beginning of the load step on all the matrices, with breakthrough reaching 10% of the HMW binding capacity in the order of the monolith, membrane, SPSFF

resin, and Poros 50HS resin. Poros 50HS exhibited significantly greater capacity for binding MAb2 HMW. For MAb1, >50% HMW breakthrough was observed at the start of MAb breakthrough for SPSFF and the Monolith SO_3 . Conversely, <20% HMW breakthrough was observed with Mustang S and Poros 50HS when MAb breakthrough was observed. The HMWs breakthrough curve on Mustang S increased sharply compared to the curve for the Poros 50HS resin. This implies that the HMWs readily accessed the charged ligands on the membrane, which had efficient mass transport; however, the binding capacity for HMWs was low. The HMW breakthrough curve increased at a slower rate on Poros 50HS and exhibited a higher HMW binding capacity. The HMW breakthrough curves of the SPSFF and Poros 50HS resins represented a typical comparison of patterns obtained from a diffusive and perfusive resin particle packed column, respectively. Breakthrough started earlier with the diffusive resin, but increased slower than with the perfusive resin during the load step. This implied that the charged ligands on the diffusive resin particle surface were available; however, due to less effective mass transport in the column, a lower ligand amount was used per unit quantity of MAb that passed through the column, compared to the perfusive resin, as well as the membrane and monolith matrices.

The different HMW breakthrough curves obtained from the membrane, monolith, SPSFF resin, and Poros 50HS resin were translated into plots of accumulated HMW amount bound on the matrices as shown in Fig. 5, for both MAb1 and MAb2. The binding capacity for the membrane and monolith was saturated earlier than SPSFF and Poros 50HS. The HMW amount bound on Poros 50HS resin increased steadily while overloading both MAb1 and MAb2. The effectiveness of SPSFF resin at HMW capture was observed after collecting 250 g/L of MAb, where the HMW amount bound on SPSFF resin surpassed the amount bound on Mustang S membrane. The accumulated HMW binding capacity ranking (Poros 50HS > SPSFF > Mustang S > Monolith SO_3) held true for both MAb1 and MAb2 using the isocratic overloaded conditions. The higher a matrix's binding capacity is for a MAb, the higher its binding capacity to be for the HMWs of that MAb. An explanation for the difference between the two MABs was that the HMW DBC for MAb2 was higher than MAb1, on all the matrices (Fig. 4). The difference of total surface charge or net surface charge of HMW species may be

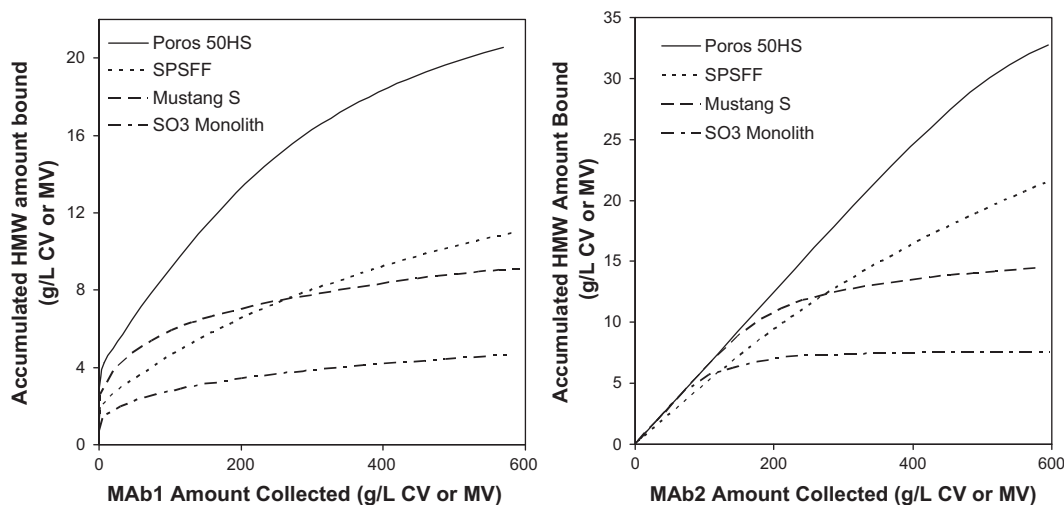


Fig. 5. Accumulated HMW amount bound from Mab1 and Mab2 on the matrices using overloaded condition at pH 5.5 and 5 mS/cm.

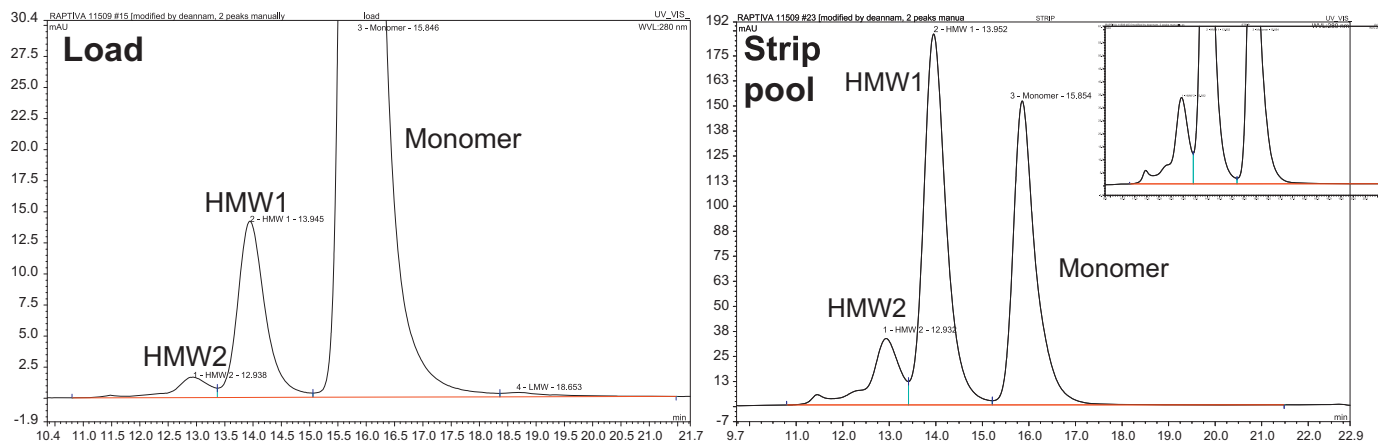


Fig. 6. Mab2 HMW species in the load and in the strip pool. The total HMW% (sum of HMW1% and HMW2%) in the load and elution pool were 6.3% and 60% respectively. HMW 1 and HMW2 peaks are mainly dimer and higher order aggregates, respectively.

a contributing factor, as the monomer binding capacity for Mab2 was slightly higher than Mab1. The pI of Mab1 is 8.2, while Mab2 has a pI of 9.0.

The total amount of HMWs were similar between the two MAb, and therefore the amount of HMW species does not seem to play a role in the binding capacity of HMW, for these two MAb. However, further evaluation into the type of HMW species was important to determine the separation of HMW and monomer. The HMW species included both dimer and higher order aggregates. Size exclusion chromatography was performed on both the Poros 50HS load and strip pool for Mab2, and two HMW peaks, referred to as HMW1 and HMW2 were seen eluting prior to the monomer (Fig. 6). The HMW1 peak primarily contained dimer, and the majority of the HMW2 peak consisted of higher order aggregates. Both peaks were enriched in the strip pool, compared to their presence in the load solution. Both forms of HMW were effectively captured during the isocratic overloaded chromatography run when Mab2 was loaded to 600 g/L on the Poros 50HS resin.

Looking further into the dimer and higher order aggregate removal on these matrices, it was discovered that the effective removal of dimer and higher order aggregates were different among the matrices tested in this study. Fig. 7 illustrates the breakthrough curves of HMW1 and HMW2 for Mab2. The aggregates normally have a higher number of surface charges compared to the dimer, and the dimer more than the monomer, but other factors

may have impacted the binding capacity to cause the differences observed among the matrices, including molecule surface charge number, steric hindrance, exclusion phenomena, mass transport efficiency, matrix ligand density and accessibility. The net result

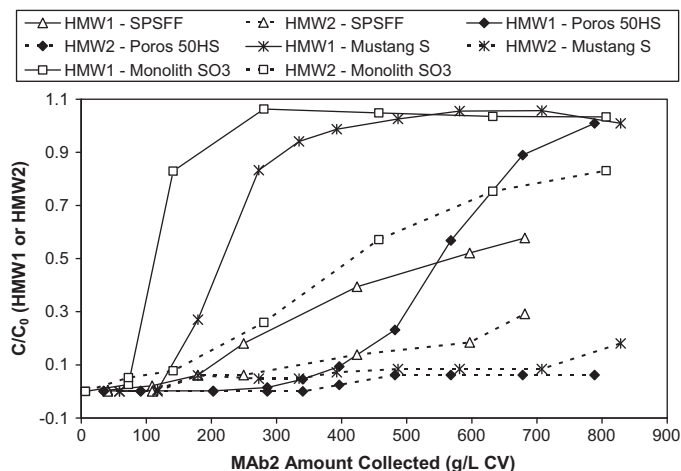


Fig. 7. Breakthrough curves of HMW1 and HMW2 of Mab2 obtained from monolith SO₃, Mustang S, SPSFF column, and Poros 50HS column.

is that the breakthrough curves appeared in the order of the monomer, the dimer, followed by the higher order aggregates for all of the matrices. The HMW species removal for MAb1 exhibited a similar trend (data set not shown). With the convective flow matrices, the HMW2 species for both MAb1 and MAb2 had a higher binding capacity on the membrane than the monolith under the tested condition. It also had a higher binding capacity on Poros 50HS compared to SPSFF resin. Poros 50HS provided the highest percentage of HMW retention among the matrices tested; this was thought to be a result of a higher binding capacity than the convective matrices and a better mass transport than the diffusive, conventional resin.

3.3. Further exploration of overloaded chromatography using conventional matrices

3.3.1. Mechanism investigation of overloaded chromatography

Experiments were performed with MAb1 to investigate whether the impurity removal was a result of displacing MAb monomer by impurities. MAb1 was loaded onto the Poros 50HS column three separate times to reach 40%, 73%, and 88% of HMW breakthrough, respectively. As load density increased, the MAb yield increased, but the total percentage of HMW removal decreased (Table 2). Additionally, the amount of MAb recovered in the strip pool (expressed in the amount bound per column volume, g/L) decreased with an increase in load, perhaps suggesting that after the point of MAb breakthrough and saturation of the column, the load material was slowly removing MAb from the column as it was displaced by another species. Further supporting this hypothesis, the amount of HMW bound onto the resin increased as the load volume was increased, as measured in the eluate pool. However, the HMWs also broke through the column at the beginning of the loading step, which would not be expected in the case for classic displacement chromatography. Thus, it is unlikely that classic displacement is occurring, but competitive binding or binding efficiency differences between species may be the dominating factor at play.

3.3.2. Impact of flow rate on SPSFF and Poros 50HS resin performance

Based on the interesting, and somewhat unexpected, observation that the perfusive resin Poros 50HS used in the isocratic overloaded mode provided considerable removal of many impurities traditionally targeted by conventional cation exchange chromatography, further work was performed to explore the potential that overloaded mode offered. It is well known that the flow rate has little effect on separation performance with convective chromatographic matrices. However, the conventional matrices could behave differently and give distinctive performance depending on the flow rate range tested and matrix property. A reduction in MAb1 DNA and HMW removal was observed on SPSFF resin with increasing flow rate (Fig. 8). CHOP removal was not affected within the flow rate range of 9–36 CV/h (data set not shown). In contrast, flowrate (18–36 CV/h) did not significantly affect the removal of DNA, HMW, or CHOP on Poros 50HS resin (data set not shown here). This could be attributed to the convective flow within the Poros resin particle and by the smaller particle size.

3.3.3. Impact of conductivity on Poros 50HS resin performance

Further studies were conducted to look at the effect of loading conductivity on Poros 50HS performance with MAb1. There was little difference in the breakthrough point when MAb1 was loaded at conductivities of 3, 5 and 8 mS/cm, but lower dynamic binding capacities were obtained when the load conductivities were increased to 13 and 18 mS/cm (data set not shown here).

Table 2
Results obtained from overloaded chromatography using MAb1 on Poros 50HS resin at pH 5.5 and conductivity of 5 mS/cm. The HMW% in load was 4.8%.

Amount loaded (g/L CV)	Breakthrough of HMW (%)	HMW in FT pool (%)	HMW bound on column (g/L CV)	HMW removal (%)	HMW1 in strip pool (P1) (%)	HMW2 in strip pool (P1) (%)	HMW in strip pool (P1) (g/L CV)	Monomer in strip pool (P1) (g/L CV)	Product recovery yield (%)
250	40	1.4	8.5	72	11.1	4.8	8.8	46.2	78.0
523	73	2.4	12.1	49	14.3	8.3	12.8	44.0	89.2
800	88	3.0	14.1	37	15.3	10.9	14.8	42.6	92.8

* HMW bound on column was calculated based on the HMW% in FT pool and product loaded amount.

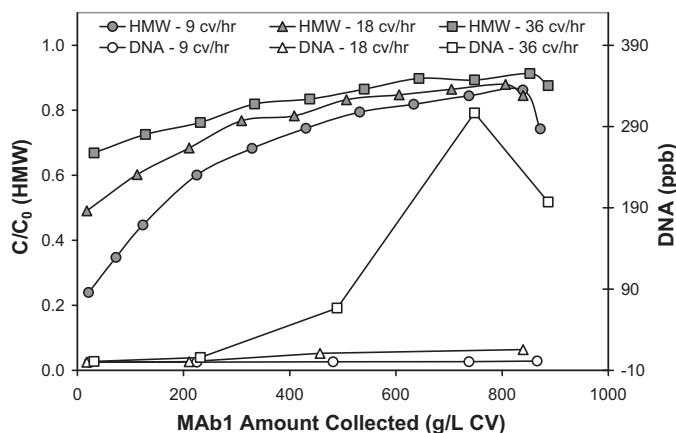


Fig. 8. Flow rate impact on DNA and HMW removal on SPSFF column. Left side Y-axis: HMW breakthrough curves, Right side Y-axis: DNA amount in fraction collected.

Several differences were observed in the chromatograms obtained during the wash, strip, and cleaning steps from the runs performed at these loading conductivities (shown as supplementary figure). In the overloaded loading condition at the conductivities of 5 and 8 mS/cm, the UV traces of the chromatograms during the wash steps decreased sharply after the wash volume reached about one column volume. Significant tailing was observed at 13 mS/cm, similar to partial binding behavior. The UV trace of the wash step at 18 mS/cm appeared as a normal flow through mode. A strip peak P1 with 350 mM sodium acetate as the strip buffer was seen for the runs at 5, 8, and 13 mS/cm, but not 18 mS/cm. The condition of loading conductivity at 18 mS/cm appears to be the least effective in retaining impurities.

The HMWs break through did not follow the same trend as previously described for MAb monomer (Fig. 9). For the runs with MAb1 in which the MAb1 was collected to 300 g/L of column volume, the C/C₀ ratio of HMW was higher at 3 mS/cm than at 5 and 8 mS/cm. It was perhaps due to the exclusion phenomena of HMW on the cation exchanger at the lower conductivity. Interestingly, the C/C₀ ratio was quite low at the beginning of the loading step and increased sharply at 13 mS/cm. This could be due to lessened exclusion effect at 13 mS/cm versus 3 mS/cm; however, binding strength would be reduced at 13 mS/cm due to elevated salt concentration in the load, and that could explain the steeper break through curve. When the conductivity was 18 mS/cm, the resin lost its ability to

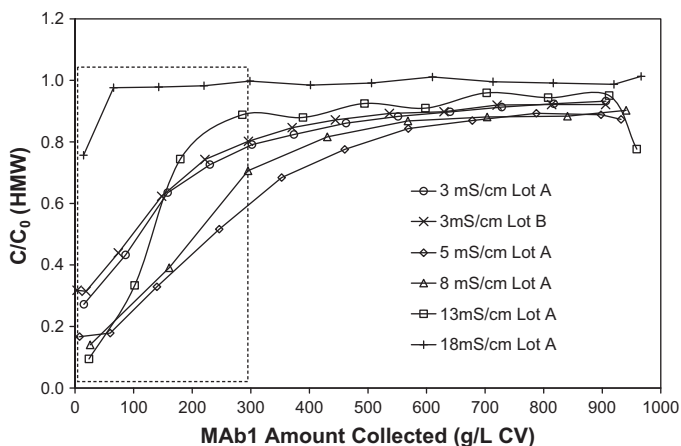


Fig. 9. Comparison of MAb1 HMW breakthrough curves obtained within the overloaded runs at pH 5.5 and conductivities of 3, 5, 8, 13, and 18 mS/cm. The runs were performed on Poros 50HS column (6.6 mm ID × 50 mm bed height).

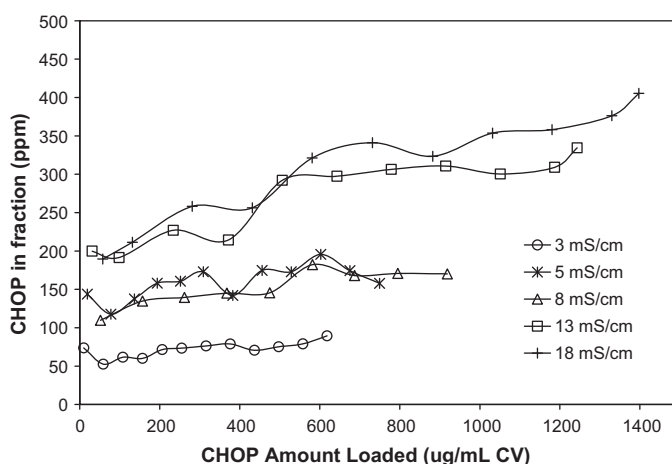


Fig. 10. Comparison of CHOP removal within the overloaded runs with MAb1 at pH 5.5 and conductivities of 3, 5, 8, 13, and 18 mS/cm, respectively. (a) CHOP amount in fraction collected and (b) HMW breakthrough curves. The CHOP in the loading pool were 682 ppm at 3 mS/cm, 846 at 5 mS/cm, 976 ppm at 8 mS/cm, 1302 ppm at 13 mS/cm, and 1446 ppm at 18 mS/cm.

bind HMWs. Therefore, to employ the overloaded cation exchange chromatography to remove HMWs, this data suggests it is important that the loading condition be optimized to maximize the HMW DBC.

The load samples were prepared through addition of sodium acetate to reach the desired conductivity, and then filtered through 0.2 μm filter to remove particles in the pool and reduce the pool turbidity. This preparation method resulted in varied CHOP content (in units of ng/mg (ppm) of MAb) in the load pool at different conductivities. The CHOP solubility increased when the conductivity was higher within the conductivity range tested in this study. The CHOP in the loading pools were measured to be 682 ppm at 3 mS/cm, 846 at 5 mS/cm, 976 ppm at 8 mS/cm, 1302 ppm at 13 mS/cm, and 1446 ppm at 18 mS/cm. Fig. 10 shows that the CHOP content (ppm) in the flow through fractions is lower when the loading conductivity is lower. CHOP removal was relatively robust within the conductivity range tested.

3.3.4. More exploration on impurity removal and charge variant distribution on Poros 50HS resin

Understanding the capability of cation exchange chromatography using isocratic overloaded mode can help design future MAb purification processes. To this end, an isocratic overloaded run was performed with a protein A pool (MAb3) as the load pool containing a higher level of CHOP (~10,000 ppm) and more than 90% of CHOP was removed during the loading and wash steps (data set not shown here).

Gentamicin is an aminoglycoside antibiotic used in mammalian cell culture processes to avoid bacterial contamination. Leached protein A is an impurity introduced into a MAb purification process when protein A affinity resin is used. These impurities can be reduced to levels below detection limits using traditional bind-elute cation exchange chromatography. MAb1 was used to evaluate the capability of removing these impurities on Poros 50HS resin under isocratic overloaded conditions. Both gentamicin and protein A leachate bound strongly to the resin (Table 3). Only protein A leachate breakthrough was observed when loading >800 g of Mab per liter of column volume. Further studies will be conducted to facilitate understanding on the mechanism of their binding behaviors.

Depending on elution peak collection criteria, MAb charge variant distribution can be altered during a cation exchange run in a traditional bind-elute mode because acidic species elute earlier

Table 3

Leached protein A and gentamicin removal from MAb1 on Poros 50HS column with overloaded condition at 5 mS/cm and pH 5.5.

Sample	Protein A (ppm)	Gentamicin (ppb)
Load	4.8	7.15
Product collected (g/L CV)		
65	<2.2	<0.08
331	<2.2	<0.08
517	<2.2	Not tested
727	<2.2	<0.08
860	3	<0.08
Strip peak	82	20

than main and basic species. The distribution of MAb1 charge variant was examined on the Poros 50HS resin under the isocratic overloaded condition in this study. Similar to bind-elute mode, the percentage of acidic species was higher in the earlier flow-through fractions, while later fractions contained greater concentrations of basic variants (Table 4). It is expected that the charge variant distribution, in the collected MAb pool, was not be significantly affected by operating in isocratic overloaded mode. Significant changes could occur if loading to a low capacity (<330 g/L for MAb1).

3.4. Incorporation of overloaded cation exchange chromatography in MAb purification

Once the high capacity and selectivity of isocratic overloaded cation exchange chromatography were discovered on the Poros 50HS column, the next step was to evaluate the entire monoclonal antibody purification process with isocratic overloaded cation exchange chromatography, without compromising product purity or recovery. MAb2 was used for this exercise.

The MAb2 purification process is a typical platform antibody process, consisting of Protein A affinity chromatography, followed by the SPSFF cation exchange chromatography operated in bind-elute mode and Q-Sepharose Fast Flow (QSFF) anion exchange chromatography in flow-through mode. Some operating conditions of the process are summarized in Table 5. An alternative process to purify Mab2 by replacing the cation exchange chromatography in bind-elute mode with the overloaded cation exchange chromatography was evaluated. As described in Table 5, the loading density on the Poros 50HS column, using isocratic overloaded conditions at pH 5.5 and conductivity of 5 mS/cm reached 600 g of Mab2 per liter of column volume, which was at least 15-fold higher than the bind-elute mode process. The loading condition for the subsequent QSFF process in flow-through mode was at pH 8.0 and 5 mS/cm. A simple pH adjustment of the MAb2 pool collected from the overloaded run was performed with Tris base solution, and there was no dilution required to prepare the load with the pH-adjusted pool for the QSFF process.

MAb2 yield and impurity clearance for the current commercial process and the alternative process were compared and listed in Table 6. Comparable step yields for the two processes were obtained. The processes also demonstrated comparable CHOP and

Table 4

MAb1 charged variant distribution throughout the overloaded run on Poros 50HS column at 5 mS/cm and pH 5.5.

Sample	Acidic species (%)	Main species (%)	Basic species (%)
Load	28.5	60.9	10.6
Product Collected (g/L CV)			
65	32.9	62.7	4.4
331	29.8	59.2	11.0
727	29.5	58.1	12.4
833	29.6	57.6	12.8
Strip peak	8.4	32.4	59.2

Table 5
Purification conditions used for current commercial process and alternative process using overloaded condition for MAb2.

Purification step	Commercial process			Development process with overloaded CEX		
	Resin	Operating mode	Operating condition	Load density (g/L CV) ^a	Operating mode	Operating condition
Protein A CEX AEX	Prosep vA	B/E	STD	20	B/E	STD
	SPSFF	B/E	pH 5.5, load 4.7 mS/cm, elution 13.25 mS/cm	40	Overloaded	pH 5.5, 5 mS/cm
	QSFF	FT	pH 8.0, 6.9 mS/cm	70	FT	pH 8.0, 5 mS/cm

^a CV is Protein A, CEX, or AEX column volume.

Table 6
MAb2 purification step yield and impurity clearance for the current commercial process and alternative process using overloaded cation exchange chromatography.

Purification step	Commercial process			Development process with overloaded CEX		
	CHOP (ppm)	Monomer (%)	Yield (%)	CHOP (ppm)	Monomer (%)	Yield (%)
Protein A	677–939	94.7–95.9	>98	995	93.4	>98
CEX	116–183	99.3–99.8	87	158	99.7	90
AEX	<1.48*	99.7–99.8	>98	2	99.7	95

* Data obtained from concentrated bulk.

HMWs clearance. Other impurities such as DNA, leached Protein A, and gentamicin were removed to below the detection limits (data not shown here) for both the processes.

4. Conclusion

In summary, the findings reported above demonstrate that both the convective and diffusive cation exchange chromatographic matrices can be operated using isocratic overloaded conditions to reduce CHOP, DNA, HMW, leached protein A, and gentamicin, while allowing monoclonal antibody to flow through the matrices after the antibody reaches its DBC. Contrasting the different matrices when used in the overloaded mode, the conventional perfusive cation exchange matrix provides a higher binding capacity than the convective matrices and better mass transport than the diffusive matrix; therefore, it provides a higher capacity to retain HMW, and removes other process-related impurities.

Cation exchange chromatography is typically operated in bind-elute mode with capacities <100 g/L. This step can be a bottleneck in downstream processing of high product titer in cell culture fluids. Using isocratic overloaded mode, instead of the classic bind-elute mode for cation exchange chromatography, opens the door to many time and/or cost saving options. If a facility is being built or retrofitted, this option allows for a smaller cation exchange column (up to 90% reduction) to be used in large scale MAb purification process, which translates to a considerable savings in both capital and materials. This method of cation exchange chromatography also allows for an existing facility that would otherwise be limited by the size of the cation exchange column or the need to perform multiple cycles to take on a significantly higher titer process. Since fewer steps and a smaller volume of buffer are needed for overloaded processing, buffer tank capacity also becomes less of an issue.

Another potential benefit of this approach is its use with disposable, pre-packed columns. Two of the major hindrances to the use of disposables in downstream processing is that (1) the pre-packed columns are too small to be used at the typical commercial scale and (2) the cost of the resin dominates the overall cost, making them impractical for uses where the resin would typically be unpacked and re-used for multiple products (e.g., pilot scale). In the case of commercial manufacturing where the resin could be used up to the limit of its lifetime, a 90% reduction in the required column volume could bring pre-packed columns into consideration, where the avoidance of packing and testing steps could lead to efficiency improvements. In the case of smaller scale applications, a dramatic reduction in the resin required, and thus the cost, makes the option of disposing of the column after only a few cycles more palatable.

Finally, the use of cation exchange chromatography in the overloaded mode could enable a more streamlined downstream process through the coupling of multiple steps together. By simply incorporating in-line pH adjustment capabilities it is possible to run a continuous cation exchange → anion exchange chromatography or mixed mode → viral filtration step. In so doing, the need for pumping skids and in-process pool holding tanks is reduced, which

lowers the required facility footprint and processing time. This also provides an option for molecules with in-process pool stability issues.

The application of overloaded chromatography to the cation exchange step using a perfusive resin and strong binding conditions enables the cation exchange step to be run in essentially a flow-through mode to garner cost and processing efficiency improvements, while still achieving the bulk of the impurity removal expected of traditional bind-elute cation exchange chromatography.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.chroma.2011.08.008](https://doi.org/10.1016/j.chroma.2011.08.008).

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